

- [Charts: visualizing data](#)
- [Getting started with BigQuery](#)

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the [full course website](#) for more.

- [Intro to Pandas DataFrame](#)
- [Intro to RAPIDS cuDF to accelerate pandas](#)
- [Getting Started with cuML's accelerator mode](#)

Using Accelerated Hardware

- [Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API](#)
- [Train a Vision Transformer \(ViT\) for image classification with JAX](#)
- [Text classification with a transformer language model using JAX](#)

✓ Featured examples

- [Train a miniGPT language model with JAX AI Stack](#)
- [LoRA/QLoRA finetuning for LLM using Tunix](#)
- [Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA](#)
- [Loading Hugging Face Transformers Checkpoints](#)
- [8-bit Integer Quantization in Keras](#)
- [Float8 training and inference with a simple Transformer model](#)
- [Pretraining a Transformer from scratch with KerasHub](#)
- [Simple MNIST convnet](#)
- [Image classification from scratch using Keras 3](#)
- [Image Classification with KerasHub](#)

```
import pandas as pd

# Sample DataFrame
order_data = pd.DataFrame({
    "Customer_ID": [101, 102, 101, 103, 102, 101],
    "Order_Date": ["2024-01-10", "2024-01-12", "2024-01-15",
                  "2024-01-18", "2024-01-20", "2024-01-25"],
    "Product_Name": ["Laptop", "Mouse", "Keyboard",
                    "Laptop", "Keyboard", "Mouse"],
    "Order_Quantity": [1, 2, 1, 3, 2, 4]
})

# Convert Order_Date to datetime
order_data["Order_Date"] = pd.to_datetime(order_data["Order_Date"])

# 1. Total number of orders made by each customer
orders_per_customer = order_data.groupby("Customer_ID").size()
print("Total Orders by Each Customer:")
print(orders_per_customer)

# 2. Average order quantity for each product
avg_quantity = order_data.groupby("Product_Name")["Order_Quantity"].mean()
print("\nAverage Order Quantity for Each Product:")
print(avg_quantity)

# 3. Earliest and latest order dates
earliest_date = order_data["Order_Date"].min()
latest_date = order_data["Order_Date"].max()

print("\nEarliest Order Date:", earliest_date)
print("Latest Order Date:", latest_date)
```

```
Total Orders by Each Customer:
Customer_ID
101         3
102         2
103         1
```

```
dtype: int64
```

```
Average Order Quantity for Each Product:
```

```
Product_Name
```

```
Keyboard      1.5
```

```
Laptop        2.0
```

```
Mouse          3.0
```

```
Name: Order_Quantity, dtype: float64
```

```
Earliest Order Date: 2024-01-10 00:00:00
```

```
Latest Order Date: 2024-01-25 00:00:00
```